$U = 314.16$ m/s $V_a = 202.35$ m/s

$V_a/U = 0.644$ $\left.\frac{\Delta h_o}{U^2}\right|_{stage} = 0.3082$

$\Delta h_o = 30.418$ kJ/kg $= W = U\,\Delta V_w$

$\Delta V_w = 96.824$ m/s

Degree of reaction $R = \dfrac{p_m - p_i}{p_e - p_i}$

$$R = \frac{h_m - h_i}{h_e - h_i} \qquad Tds = dh - vdp$$

$$h_i + \frac{W_i^2}{2} = h_m + \frac{W_m^2}{2}$$

$$\frac{\gamma}{\gamma - 1}\frac{p_i}{\rho_i} + \frac{W_i^2}{2} = \frac{\gamma}{\gamma - 1}\frac{p_m}{\rho_m} + \frac{W_m^2}{2}$$